Who Watches The Watchmen?

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Every story has a beginning, a middle, and an end. The difficulty in faithfully recounting a story detailing a rapid succession of events that individuals lived th is that it is a tale told in a thousand ways: a collective dream that meshes togethe blending facts with fiction, living in the never-never land of symbols and portent And, as is the case with dreams, sometimes there is a monster at the end of it.

I.
Les Prophéties

My story begins on May 02, 2026.

While idly scrolling social media app X (formerly Twitter), I come across a tweet Will Depue, previously an employee at the American superintelligence corporati OpenAI. The tweet says (verbatim) “we’re really living through a historic momen mathematics. it’s pretty incredible to watch”. Reading the ensuing conversations mathematician Daniel Litt in the replies, it dawns on me that the tweet alludes t future impact of LLMs in the field of mathematics.

I must admit that at this point of time I had not really been following the developments happening in the space between mathematics and AI, despite bein mathematician myself, although occasional news about progress (eg, on Olympia problems, various Erdős problems, …) would slip through the cracks and appear c news feed. Nevertheless, I had certainly heard much about AI itself, primarily fro The Everything App, which happens to be the typical congregation spot for buzz surrounding all things AI. Much of these conversations are no doubt populated a amplified by individuals, like Depue himself, who are based in the San Francisco Area, a place that is intimately tied to the genetics of the AI boom.

Much has been said about how this unending 24/7 AI-news cycle originating fro
Bay Area has to do with Silicon Valley VC startup funds being intimately tied to
great financial pyramid scheme of the century, working tirelessly to ensure that t
capex breathes while starving every other aspect of the palpable economy and gl
financial ecosystem of oxygen, all at the cost of leaving world economy hanging i
balance, reliant on speculative fortunes of the One Bet. What I had not realized, a
could not dream of, was how my relatively quiet and monastic inner life as a
mathematician would soon be dragged into the whirlpool that is "Project AI".

The discussions between Depue and Litt went back-and-forth. Litt firmly made t
point that, even though solving problems is a significant part of what mathemati
do, the discipline of mathematics is strictly less directionally bound by the kind c
precise questions that one can train AI systems on. On the other hand, Depue ke
insisting, almost as if he's reaffirming his unshakable blind faith in the second co
of Christ in the form of a Machine God, that the same framework of RL that exce
at solving the Olympiad problems will (verbatim) "more slowly solve the rest [of
mathematics]".

At this point, an unlikely character enters the conversation with a response to Li
rebuttals: Jacob Tsimerman, who would eventually go on to win the Fields medal
few months. He was already known within mathematical circles to be one of the
possible contenders for the Fields medal this year. I'd like to pause here and let m
readers know that at this point of time Tsimerman himself already knew that he
secured the Fields medal; this news was conveyed to him in January by the presid
the IMU.

Continuing with the story: Tsimerman responds to Litt by construing the domai
mathematical investigations as one that, eventually, centres around solving speci
problems anyway, once they are formulated precisely. Therefore, he argued, can c
really say with certainty that the impact of AI in the field would not be so signifi
and severe?

On May 04, Tsimerman goes on to elaborate on his points by a lengthy tweet.

He makes the following claim (verbatim): “problem-solving is an immense, and pervasive part of modern mathematical research. Consequently, if human proble solving disappears by virtue of the AIs becoming strictly and substantially better then most of the time currently spent by modern mathematical researchers will h to be spent on an activity that is altogether pretty different. Whether such an act is viable as a professional endeavour is something I am unsure of …” Overall, the of this post seemed rather urgent. Tsimerman spoke like Nostradamus warning c undeniable impending apocalypse that he had in his visions.

From my vantage, the discourse was rather confusing. Having grown up in the pedigree of Henri Poincaré and Bill Thurston, the claim presented by Depue and Tsimerman that large swaths of mathematics ultimately boils down to problem-solving seemed misplaced at best, and disingenuous at worst. Whether mathema problem-solving or not always seemed more a matter of taste than an intrinsic question about the nature of mathematics, to me. But more importantly: there we major news of AI solving major mathematical problems of general interest at thi point of time. So what on Earth were these people even talking about?

I remember my friend, who is also a mathematician, sent me the humorously wor message: “how can a serious mathematician say this “my entire main research fie disappears if someone solves the *Zilber-Pink Conjecture* in its more general form”? quoting (verbatim) a line from Tsimerman’s tweet.

II.
Start Of Something Beautiful

On May 20, two and a half weeks after the exchanges between Depue, Litt and Tsimerman, OpenAI announced that their internal models had disproved the “un distance conjecture”, posed by Paul Erdős in 1946, (rightly) described in their mil research announcement as “one of the best-known questions in combinatorial geometry, easy to state and remarkably difficult to resolve”.

At the time, it felt like our Nostradamus’ predictions are coming true before our eyes. I would eventually come to realize that these predictions were not proclama

of a prophet, nor paranoid ramblings of a delinquent, but something more sinist
smokescreen— a seemingly plausible reframing of the essence of mathematics a
impact on the real world, something that our generally naive and narrow-minded
intellectuals could have sterile debates over, while unnoticed behind the curtains
certain entities could continue their project of large-scale institutional capture a
economic disenfranchisation of vast sections of human society.

The graphic novel *Watchmen*, originally a 12-issue comic book series, was written
the 80s by the self-proclaimed English wizard Alan Moore and is now hailed as o
the most influential comic book literature to date. The central theme of *Watchme*
meditation on the concept of power. The book is set in the backdrop of an alterna
80s America, where US won the Vietnam war, Richard Nixon is still the presiden
the Cold War rages on. The ones responsible for this divergence in history are th
protagonists of the novel, the Watchmen collective, entailing various anti-hero
vigilante characters, each with their own conception of power and flawed visions
morality and that of greater good. The main plot of *Watchmen* follows several of t
characters as they slowly uncover their comrade Ozymandias' plan to unite the w
and avert the impending nuclear armageddon by sacrificing millions of lives and
blaming it on a fake alien invasion. At its core, *Watchmen* is a story about the vice
the American society. The world of *Watchmen* is bleak and violent, seen through
overbearingly nihilistic lens of the masked vigilante Rorschach as he roams the c
ridden streets and backalleys of a gothic-noir version of New York City. Through
the novel the reader is catapulted across a jarring barrage of newspaper cut-ups,
television media, journals, memoirs, stories-within-stories that emphasize a colla
of the information ecosystem not unlike the present world we live in. In face of th
existential crises, the heroes of the novel retreat into sex, drugs, religion, delusio
anything to avoid reality. *Watchmen* posits that if a superintelligent omniscient b
really did exist, the American government would no sooner reduce it to a weapon
a propaganda tool.

The research announcement by OpenAI on the unit distance conjecture is
accompanied by a short teaser launch video. It is here that we are introduced to s
of our proverbial Watchmen: visionary AI researchers at the frontier of Mathema

and AI, witnessing the "superintelligence" emerge at their terminal as out of not suddenly realizing that they are the ones upon which a greater part of the responsibility is bestowed to chart out a course forward for humanity into the Gr Unknown.

Cinematic music reminiscent of Hans Zimmer's compositions plays in the background. We see OpenAI researchers, previously career mathematicians, Mar Sellke and Mehtaab Sawhney, explaining the unit distance conjecture on an easel chalkboard; camera pans revealing quick flashes of interior design boasting Silic Valley startup decor. Sellke and Sawhney both recall their initial disbelief upon s the model output; Sellke recounts having trouble sleeping upon witnessing the p Next, we are introduced to Ozymandias himself: then-head of the OpenAI math research team, Sébastien Bubeck. Sébastien succinctly explains how AI succeede humans here: the proof-tree for this approach was a dark-forest, there were too n delicate decisions to make; by searching through all possible approaches, the relentless blind idiot god Yaldabaoth had found the right idea, like a lightning st the ground. He ends with the following remarkable comment (verbatim): "And so this tells us, is that AI is able to make breakthroughs in science, and can enable breakthroughs not only in mathematics, but in engineering, in physics, in biology medicine … (pensively) This result is showing glimmers of a Golden Era for mathematics." At the beginning and the end of the video, Fields medalist Timoth Gowers also makes quick appearances, stating rather matter-of-factly, in his usua British demeanour, the importance of the result that has been proved by AI.

I am far from a combinatorist, but I had heard of the unit distance conjecture bef as it closely relates to the distinct distance problem by Erdős which has been bot source and part of much interesting mathematics. Notably, work due to Larry Gu and Nets Katz on the distinct distance problem used strikingly simple new ideas polynomial methods and real algebraic geometry. While I did not find, admittedl upon rather cursory reading, OpenAI's internal model's proof of the unit distanc conjecture nearly as gripping as the work of Guth-Katz on the distinct distance problem, it was clear that proof had some decently interesting ideas extending E original number-theoretic constructions. Furthermore, the experts seemed pretty

much unanimous that this was an exciting development. The paper produced by model was decently written, and it was accompanied by a paper co-authored by s experts summarizing the ideas of the proof very clearly, together with additional commentary by each of the experts — Tsimerman's comments began: "This is a r impressive piece of work, and I would accept it for any journal without hesitatio Bubeck's prediction of the future seemed on the horizon: a Golden Era of mathematics, fuelled by a synergistic collaboration between humans and AI, is n upon us.

Everything flows. All nice things come to an end. The unit distance conjecture w mark the last time OpenAI collaborates with the broader mathematical commun good faith. Since the initial highs, the story of AI corporations' obsession with mathematical theorems has increasingly felt like coming down an up-staircase… the encounter at the end was crushing.

III.
The Men Who Sold The World

In the weeks following, the discourse on X seemed to return time and again to th topic of mathematics. Comments of the form "Can AI run out of math problems solve?" were abound. Everyone seemed to be engaging with a performative caric of mathematics through the lens of AI. At this time, academics around the world started slowly realizing that GPT-5.5 is able to, once in a while, generate nontriv mathematics upon prompting sufficiently many requests. It seemed this news ha percolated to X. Every article on arXiv with a morsel of contribution from AI wo get posted, screengrabbed and circulated across X, as if heralding the arrival of a unseen superintelligence.

The next breakthrough came on July 19th from an innocuous tweet by Anthropic employee, previously a career mathematician, Levent Alpöge. The tweet said (verbatim): "hello there the jacobian conjecture is false thanx to my close friend a for asking about it and my other close friend fable for working during the world c final", followed by a string of symbols giving a three-component polynomial map three variables that disproves the Jacobian conjecture in dimension 3, a problem

dates back to 1939. The formula could be checked on any old computer algebra sy
and upon doing so one would soon conclude that it is indeed a valid counterexam

I remember the tweet appearing on my feed when it had less than 50 likes. I was
familiar with Alpöge prior to this, so I had no idea if the author was reliable, and
wondered for a while if it was a prank of sorts. Slowly, everyone including me car
the realization that it is indeed correct. The news spread like wildfire among the
netizens of X. My initial reaction upon verifying the solution was… disappointme
have always thought that an answer to a mathematical problem that is resolved b
collection of symbols without real insight is rather boring, regardless of how old
prestigious the problem is. At the time, I also thought that this was consensus op
among mathematicians; I now realize that this was a misjudgement on my part. T
original tweet by Alpöge came with no provenance on how the answer was reach
what prompts and harnesses were used; a pattern that would repeat over and ove
the coming weeks.

Regardless, it was soon found by someone, again using AI, and then confirmed in
follow-up tweet by Alpöge, that there is a much more conceptual (indeed, geome
way of thinking about the example. The conceptual explanation nevertheless left
something crucial to be desired, an algebraic coincidence that had seemingly no
intuitive explanation. To my knowledge, no document detailing these points rega
the solution to the Jacobian conjecture has been released by Alpöge so far, despit
promising the mathematical community (verbatim) "there'll be some pdf eventua
After all, let's face it: is it really important how the solution came to be, what thr
of human mathematical ideas, what corners of the castle of knowledge and
understanding that we have painstakingly built brick-by-brick for centuries, that
Anthropic's internal model "Fable 5" trawled through, to get to the solution to th
conjecture? No, obviously not… the easy-to-state and truthful headline is machin
solved an old prestigious conjecture that humans did not. What more could you
possibly want to know?

The resolution to the Jacobian conjecture, studied by countless mathematicians
(Zariski, Miyanishi, Gurjar, Abhyankar, Moh, …) lies gathering dust in a 280 chara

tweet with 43K likes, authored by Anthropic employee Levent Alpöge on the 19t
July, 2026. Much has happened since, and who has time to dwell in the past?

Friday 27th August 1999, Sheraton Moriah Hotel, Dead Sea, Israel. A discussion
geometry is being held, moderated by Misha Gromov, as part of the "Visions of
Mathematics: Towards 2000" conference hosted by the Tel-Aviv University. I hig
a conversation from near the end of the discussion:

> P. SARNAK: I think one point that is coming up here, and it came up in your [Gromov's] discussion, is, and I think this is going to be very relevant in the fu is the role of computers in proofs and in mathematics, and it doesn't seem to h been discussed. It's inevitable that computers will appear in mathematics, I m computers in proofs, not in science, not in numerical analysis, the role of com in pure math, in something like the proof of the four-color theorem or the pro the sphere packing problem.
>
> M. GROMOV: I have a more radical opinion. I believe, that may be in 50 years mathematicians will be less relevant than computers.
>
> V. MILMAN: You see, Peter [Sarnak], if the Riemannian hypothesis would be s with the help of computers, I would consider this to be a great achievement, because it doesn't matter how it will be solved. When you talk about the four- problem, the only interest, which exists in such a problem, is that if this quest so simple and we don't understand how to answer it, then I would expect that very young guy will find a missing structure here, and this structure will be m important for mathematics than the four-color problem. And when you put in thousand hours of computer time just to tell me that I will be able to find the colors, indeed I don't care. If some mathematics was developed by it, that is th thing I care about. [COLLECTIVE LAUGHTER]

Peter Sarnak is the doctoral advisor of Jacob Tsimerman, and is thanked in Lever
Alpöge's doctoral thesis for years of mentorship. It is inspiriting to see the pupils
venturing out and fulfilling the teacher's dream: a future where computers have a
undeniably significant role in novel mathematical proofs. Nevertheless, one cann

help but stop and wonder: what if the dream was really a nightmare?

IV.
Portrait Of Two Mathematicians

On July 23rd, four days after the resolution of the Jacobian conjecture, at the Fiel medal ceremony, Jacob Tsimerman accepts the Fields medal, wearing a wide smi a custom-tailored silver tuxedo. Later that day, during a press conference, he announces that he would be joining OpenAI's AI safety division. He says (verbati think the world is changing … The math profession as we know it now … I don't t it will exist the way it exists right now."

It would be an understatement to construe Tsimerman's alignment towards AI research (and indeed, towards OpenAI in particular) as one more milestone in th project of institutional capture by AI corporations that has been slowly but surel taking place over the last few years. Four AI companies, Open AI, Anthropic, Go Deepmind and Meta has been consistently poaching academics across discipline They recruited more than 80 professors from top American universities in recent according to some dated statistics. The numbers have probably grown steadily si In 2024, the Nobel prize for physics and chemistry were both given in relation to research. Math has not suffered this gloomy a fate: the 2026 Fields medal was aw to Tsimerman firmly due to his human achievements in number theory. Howeve stated decision to take leave from University of Toronto to move to AI research v obviously one that was a long time coming, possibly (and this is speculation on m part) already finalized in January 2026 when he privately received confirmation to secured the Fields medal. Indeed, exactly a year ago, in 2025, Tsimerman had co-authored an article titled "A Taxonomy of Omnicidal Futures Involving Artificial Intelligence", where he lists various AI doomsday science-fiction scenarios. In ar essay, mathematician Max Weinreich states (verbatim) "In the last two years, [Tsimerman] only took graduate students interested in AI; then, he no longer too graduate students; then, he left number theory to work on AI research."

In a Quanta Magazine interview, Tsimerman says (verbatim) "In my experience, a think many people's experience, math is very much a goal-oriented endeavor … th

truth-and-beauty stuff is there when you're zoomed out, but when you're zoomed
you just want to win." The article goes on to say that, after he performed poorly i
Canadian Mathematical Olympiad at age 12, he'd spend five hours every day prac
Olympiad problems. Subsequently, he achieved a perfect score in the Internation
Mathematical Olympiad in 2004. After that, he stopped competing because he'd
achieved a perfect score and participating again could only damage his record. Ir
when he realized he could be a serious contender for the final eligibility cycle of
medal, his childhood ambitions of the Olympiad days returned; he decided to pu
single-mindedly: "I worked hard and sacrificed enjoyment to try and win it." By h
own admission, he did this by deliberately reorganizing his research area to work
more "impactful" problems, solutions to which could secure him the Fields meda

On October 16, Tsimerman announced he would be working at the Mathematica
Safety Institute (MAISI) full-time as its founding scientific director, instead of jo
OpenAI, although nevertheless they would still be collaborating with them. He a
announced that MAISI will be supported by OpenAI. After all, he who sells the p
also sells you the antidote. It is becoming increasingly clear that AI safety is bein
used as a smokescreen by AI corporations themselves to keep the intelligentsia b
with trivia while erasing the very real concerns about IP, agency and product liab
from public consciousness. On September 24, Edward Snowden, while speaking a
Ethics Forum at ETH Zurich, argued that AI corporations must bear liability for
harms caused by their products just like a weapons manufacturer or a drug
manufacturer does for their products. He concluded (verbatim): "I think we need
Sam Altman in jail, and I think it would be an instructive experience." Indeed, if
safety is of paramount concern in the near future, the only way to effectively deal
it is by holding the orchestrators behind these dangerous products accountable, r
being a snake oil salesman.

Tuesday, 16th August, 1983, Warsaw. An awkward looking man wearing a plaid s
and stretch denim jeans receives the Fields medal from Władysław Orlicz at the
opening ceremony of the International Congress of Mathematicians. The name o
man is Bill Thurston. Thurston's 1992 article "On proof and progress in mathema
would come to be one of the most prescient essays on the ecology of mathematic

a time around three decades hence, to where we are at: the here and now. In his e
Thurston recounts the ecological affects of early success in his career (verbatim):
that time, foliations had become a big center of attention among geometric
topologists, dynamical systems people, and differential geometers. I fairly rapidly
proved some dramatic theorems … It was hard to find the time to write to keep up
what I could prove, and I built up a backlog. An interesting phenomenon occurre
Within a couple of years, a dramatic evacuation of the field started to take place .
People told me (not as a complaint, but as a compliment) that I was killing the fie
Graduate students stopped studying foliations, and fairly soon, I turned to other
interests as well." Thurston gives two explanations for this phenomenon:

> First, the results I proved (as well as some important results of other people) w
> documented in a conventional, formidable mathematician's style. They depend
> heavily on readers who shared certain background and certain insights. The th
> of foliations was a young, opportunistic subfield, and the background was not
> standardized. I did not hesitate to draw on any of the mathematics I had learn
> from others. The papers I wrote did not (and could not) spend much time expl
> the background culture. They documented top-level reasoning and conclusion
> I often had achieved after much reflection and effort. I also threw out prize cry
> tidbits of insight, such as "the Godbillon-Vey invariant measures the helical w
> of a foliation", that remained mysterious to most mathematicans who read the
> This created a high entry barrier: I think many graduate students and
> mathematicians were discouraged that it was hard to learn and understand the
> proofs of key theorems.
>
> Second is the issue of what is in it for other people in the subfield. When I sta
> working on foliations, I had the conception that what people wanted was to kn
> the answers. I thought that what they sought was a collection of powerful prov
> theorems that might be applied to answer further mathematical questions. Bu
> that's only one part of the story. More than the knowledge, people want *person*
> *understanding*. And in our credit-driven system, they also want and need *theore*
> *credits.*

This was a man who had not only made breathtakingly beautiful new discoveries geometry and topology, but spent quite a lot of time understanding and caring fo ecology of mathematics. Later in his mathematical career, Thurston would overc for his perceived mistakes: he never published his 1986 preprint, "Hyperbolic Structures on 3-manifolds, II" containing his "Fields medal result": atoroidal Hal manifolds are hyperbolic. In 1998 he would put it up on arXiv as an e-print. The a abstract says (verbatim) "this eprint is based on a 1986 preprint, which was refere and accepted for publication, but which I neglected to correct and return. The re corrections have now been incorporated, but it is largely the same as the 1986 ve (which was a significant revision of a 1981 version)."

But this is not to say he did not engage with the mathematical community. He di simply dump his manuscript in a pre-internet era GitHub and vanish from the fa the Earth. Quite the opposite!

By early 1980s, Thurston's ideas regarding geometrization were already well kno the pertinent mathematical circles. In the mid-1970s, Thurston had given several explicating his ideas on classification of surface diffeomorphisms, which is a cen piece of the puzzle in his proof. He gave lectures on this at Orsay in 1976-77, whi would later be compiled in the Séminaire Orsay lecture notes "Travaux de Thurs sur les surfaces". Jean-Pierre Otal recounts in "Thurston's Hyperbolization of Ha Manifolds" that Thurston announced his hyperbolization theorem in Spring 197 the middle of his lectures on hyperbolic 3-manifolds. During 1978-80 Thurston b teaching a graduate course at Princeton, and in doing so he started preparing the lecture notes that would become "The Geometry and Topology of Three-Manifol In a 1994 autobiographical essay, he recalled that requests for these notes becam numerous that his mailing list reached approximately 1200 mathematicians, to w he sent instalments every couple of months. Seminars around the world starting working through these notes, often spending weeks understanding a section. On February 1980, Dennis Sullivan gave a Séminaire Bourbaki lecture outlining Thu hyperbolization for 3-manifolds fibering over the circle. This would be the first written account of the theorem. Thurston's own manuscript of the proof would b prepared a year later in 1981, where he would acknowledge Sullivan for persisten

scrutinizing and improving on his unwritten proof.

Thus, due to Thurston's efforts in communication and engagement with the mathematical community far and wide, his ideas would disseminate much faster a mere manuscript of the proof ever could, and breathed life into the field that otherwise would not have happened so swiftly and organically.

Recently, in an interview with Curt Jaimungal, Tsimerman proclaims (verbatim) ' think pretty soon, we are going to have proofs no one understands, and they're g to underlie mathematics, and based on that mathematics, other things are going work. Does a train run even if no one understands the mechanism by which it ru Yes it does, I mean, you can look at it run. Already now, a lot of my theorems tha have proven, I don't understand all the steps to …". Like many things Tsimerman publicly, this is an appeal to normalize engineering epistemology in mathematic solidify the narrative that AI is going to be far better at mathematics compared t human mathematicians. If the consensus definition of mathematics is radically altered, this would of course be tautologically true. I'm fairly certain that OpenA would be quite happy to mould such a vision of the future into reality.

Introduction of AI in mathematics as a vast industrialization project stands in ex contrast to Thurston's sensibilities. Thurston was an environmentalist in spirit. T him, mathematics was indistinguishable from mathematical ecology. He had understood that mowing down flora and fauna of the earth to build theorem-producing factories in their place is an affront to mathematical life itself. In my v we would do well to heed his call to cultivate mathematics rather than strip-min earth for theorems. But I fear it may already be too late. If fate frowns, we all per

V.
It's About Sending A Message

00:34 AM Pacific Time, Saturday, August 01. OpenAI announces 10 advances in mathematics and computer science, as a launching ground for their next generat model "Astra".

If you found "00:34 AM on a Saturday" to be an odd time and date to inform the

about your next major product planned for launch, you likely possess formidable business acumen. The mystery is unraveled a little bit upon reading Sébastien Bubeck's tweet (verbatim), "yes, nonsofic groups exist: this statement is one of th many new beautiful results proved by Astra, our next major model". Obviously, m than one person was working late at night that day; of course, to be clear, I say "l because that's what midnight is according to their tight sleeping schedules, caref logged in their Apple watches, meticulously following health protocols put forth Silicon Valley techno-longevity gurus.

What happened was this: approximately 12 hours prior, an anonymous account o had released leaked screenshots from a preliminary draft of one of the results pro by OpenAI's model: the existence of nonsofic groups. Independent rumors, credi otherwise, began circulating that Anthropic employee Levent Alpöge had someh also obtained the 10 problems list and was prompting Anthropic's internal mode Fable to beat OpenAI to the punch. Amidst the chaos, OpenAI had to finish writ up the paper and reasoning walkthroughs as fast as possible and put it up on the website. The shoddy job is rather transparent (and will become even more transp soon) from the "teaser trailer" associated to the research announcement page, wh half a minute long, animated hovercards only showing Problems 1, 7, 8, 9 and 10 accompanying pretty pictures. For context, the existence of nonsofic groups is Problem 3, and was by far the best mathematical result proved by OpenAI's mod far.

This is where I enter my own story.

The original announcement article by OpenAI on the existence of non-sofic grou claimed that there had been (verbatim) "no progress on the main result for at leas decade". Mathematician Francesco Fournier-Facio released an article on August significantly shortening the construction, and pointing out that this claim by Op was blatantly false, as their own work critically relied on the results from a prepr Gabor Kun, and a joint preprint by Andreas Thom and Gabor Kun, respectively. OpenAI has since retracted this claim.

On August 04, I decided to run these preprints by OpenAI's best publicly availab

model Sol, to check for errors. The reason was twofold: the preprints were unpublished, and it was conveyed to me by some human mathematicians that the were some tentative uncertainty as to their veracity (my understanding is this trepidation has since been completely quenched, as the authors and the mathem community has done a tremendous job quickly disseminating the ideas in these preprints). Sol came up with, what seemed to me at the time, to be a non-trivial (l it turns out, non-critical) error in the statement of the main theorem in Gabor Ku article. The theorem is a TFAE ("the following are equivalent") statement, entaili four statements. The proof goes (1) implies (2) implies (3) implies (4) implies (1). In proof, there was an error in the step "(4) implies (1)": this implication could not possibly be true as written. The catch was, OpenAI's article quotes this main the as-is, without any footnote pertaining to the error. This immediately raised red fl my head: the proofs of the 10 results by OpenAI are all accompanied by Lean (for verification system in mathematics) verification certificates. Surely the system w have found the error? Besides, if a weaker model like Sol can find it autonomousl without help from Lean, why can't Astra?

I wrote out the counterexample to "(4) implies (1)" provided by Sol, by myself, in c convinced myself it is correct, and then tweeted it out. The tweet got ~120K view eventually was seen by interns and employees at OpenAI.

The first response in the form of a quote-tweet came from Cambridge University undergraduate student Kevin Barreto, going by the pseudonym AcerFur, who at t time was doing an internship with OpenAI. His response (verbatim), "Anyways, w were aware of these things and of course checked the whole citation tree of resul before announcing such results", and then he proceeds to list two examples from Kun and Kun-Thom papers that Lean had flagged, and how Astra had applied var "patches" to get around these problems. In further public exchanges, upon being pressed by me and the pseudonymous account doomslide, Barreto admits that in prose proof there were footnotes for pointing issues, adding "But I don't want to about it further".

Something was amiss, and I suspected that OpenAI, having created a proof-gene

machine, was treating mathematical proofs with the same fast-and-loose "move f and break things" philosophy as Silicon Valley startup enterprises. As I am sure r recognized at the time, long term ramifications of this ideology taken to its natu limit is terrifyingly dystopian for the practice of mathematics (and, in future, ine other arenas of science and technology). Did anyone at OpenAI even understand proof? Or did they let Astra do all the work, checked if the definitions are the "co ones" in the proof tree of the formalization (which can also be relegated to an AI agent), and just dished it out?

The suspicion that this is the case grew tenfold the next day, when OpenAI empl Mehtaab Sawhney posted a tweet saying (verbatim), "We have heard concerns ab the proof of the existence of a non-sofic group, particularly its reliance on results Kun and Kun-Thom. The original lean certificate is end-to-end formalizing every necessary ingredient from those papers … Such issues are commonplace in the mathematical literature and do not affect the results we cite."

I do not think there is anything "commonplace" about a group of people writing paper using an AI model by typing a sequence of superficial prompts that amoun "please solve this problem", the output of which is then formalized (if that is ever correct order of how things operate), and claimed to be a complete work contribu to the broader mathematical literature, without *any* human in the loop understan just what the hell is going on. And if they did understand, why didn't they simply out a footnote explaining why these errors are minor and do not affect the proof? anything, such explanatory remarks *are* commonplace in mathematics, and contr positively to the collective understanding of mathematics! There is absolutely no you can slice the situation that makes OpenAI's math research time look good he

At this point, it should be obvious that OpenAI had already been framing a corpo narrative for the future of mathematics, one which has no need of humans. Of co their construction is ludicrous, because mathematics is nothing if not a human endeavour.

VI.
Augustus Horribilis

Over the course of the rest of August, several AI breakthroughs in mathematics v
be announced by Anthropic employee Levent Alpöge in increasingly nauseating
of nonchalance. The tweets came with no insight into the provenance of these re
no detailed log of the prompting techniques and harnesses used and had very litt
say about their relationship of the results with the existing mathematical literatu
Eventually, Alpöge, who is officially a pre-training researcher in Anthropic, woul
on to reveal that all but a handful of these were his personal projects, that he wou
tinker with Anthropic's internal model "Fable" on free extra tokens given to him
weekends. Nevertheless, his blasé tweets would rake up hundreds of thousands o
engagements on X and it would be unusual if that wasn't doing Anthropic's
capabilities marketing a favour.

On August 10, Anthropic announced that they have improved a bound on the den
of zeroes of the Riemann zeta function falling on the critical line. The announcer
said that a high-school drop-out and former "Thiel Fellow" Jared Sumner, who, i
emphasized, had no substantial expertise in mathematics, persistently prompted
Claude to "seriously attempt the Riemann hypothesis". From Sumner's own twee
(verbatim), "8 days ago, while jogging, I asked Claude to solve the Riemann
Hypothesis. It didn't. 1.5 days later, it proved >= 67% of the zeros are on the line (
41.6%). Still not sure what that means, but some analytic number theorists seem
excited". As reported by Anthropic, his input largely consisted of motivational
prompts in the general format of "keep going". Alpöge was one of the people invo
in reviewing the mathematics.

The message communicated was clear: "frontier" mathematics is now a spectator
sport. You can do it while watching the world cup, you can do it while jogging, yo
do it while being hammered at a jazz bar. The days of experts, careful scholarship
mathematics as an act of meditation are over. No doubt the adderall-infused San
Francisco techno-optimist hollow men are overjoyed by this news. It was an
impeccable marketing strategy. I am sure Anthropic's rivals at OpenAI were wat

In the days following, Alpöge posted several other solutions to long-standing ope
problems in mathematics in the form of cryptic viral tweets. Somewhere around

time, an unrelated human breakthrough came into the news: Simon Brendle and Ken Hung had solved the long-standing question of whether the product of two 2 spheres admits a positive sectional curvature metric, in the affirmative. Alpöge mentioned that he had been working on this with Fable, which produced some plausible-looking results in this direction. In a quote tweet, I suggested (verbatin it does work, let me suggest an idea (which you may ignore). Start a public blog ir name, totally decoupled from Anthropic. Make this your first post; this result is already proved by Brendle-Hung, so it would not be that useful for marketing any Invite expert opinion." Needless to say, my idea was ignored.

On August 23, Alpöge posted a tweet announcing a 108-page construction of a complex structure on the 6-sphere, which was known to be a famous open proble differential geometry. The paper, entirely generated by Fable, was accompanied b page neuralese summary of the construction, which appeared to also be LLM-generated. To date, I have not personally encountered a fully satisfying explanati the construction, although eventually Philip Engel wrote a much more readable account of the construction. It took several days for me, even with LLM-use, to d the human-unreadable paper, to convince myself, after drawing many pictures, th least the 6-manifold constructed is indeed the 6-sphere.

Since then, I have understood a few more things about the construction that are unsaid in the 108 page tome, and have been able to apply these to produce some limited number of examples of not-entirely-obvious complex structures on almos complex 6-manifolds. I suspect I have only barely scratched the surface in terms what else could be done with the ideas in the construction, and in large part beca no one really knows what the true thought process behind these ideas were.

The worst part is neither Alpöge nor the vast majority of mathematicians and no mathematicians on X, who made a lot of exciting noises about it, really seems to too much beyond the headline "AI can do stuff humans wanted to but couldn't". W have suddenly found ourselves at a possible cul-de-sac in the culture of science, certainly already gaining a slow but steady foothold in the public consciousness f years no thanks to a barrage of superficial popular science texts, wherein everybo

wants to know but nobody wants to understand. Even if they do want to understa the meaning of "understanding" seems to be heavily diluted— type such and such prompts to summarize and clean up the exposition and then upon a cursory or no cursory reading, go "ah, yes, I get it now". This is understanding without putting effort at finding something deeper or hidden behind the page of symbols and diagrams, a version of frictionless (thus, fast) mathematics. I can't help but suspe this superficialization (or even commodification) of human intellectual work was exactly the deliberate ultimate goal of announcing these AI generated results on moving social media site.

Eventually, Levent Alpöge would go on to poach various domain-experts for pers collaborations on prestigious problems outside of his domain of expertise. He wo reveal that on September 2025, he had emailed mathematician Tristan Buckmaste collaborate with him on the Navier-Stokes equation. The outcome of this collabo would set off a chain of events that would eventually lead to catastrophe for scien norms for professional mathematics. After getting, for all intents and purposes, scooped by OpenAI on the Navier-Stokes problem, he would lament the experien (verbatim) "dream crushing". Afterwards, he would go on to collaborate with Eth Sussman on another Millennium Prize problem, namely the 4-dimensional Yang- existence & mass gap conjecture, which is currently a work in progress.

At that time, it was not completely clear to me what exactly is gained by an ex ca mathematician of Alpöge's stature, in prompting LLMs on weekend projects to produce answers to mathematical questions that are too far outside his research (number theory) for him to fully understand, which are then stored in a locked bo with the keys thrown out, or released as hundreds of pages of neuralese mathema that no one ultimately has any motivation to read, or laundered through domain- experts thinly veiled as academic work. It seemed to me a fundamentally anti-soc act to announce these, especially as someone for whom the mathematical commu was once home.

I think I may now have a better theory of mind for such behavior. To explain my of view, I need you to indulge me in a brief digression.

VII.
Of Adam's Curse

The port city of Duluth, Minnesota overlooking Lake Superior, with thousands o of green space and pristine wilderness alongside the lake and over 300 miles of d trails for hiking and exploration, is a popular summertime travel destination. Th Duluth REU is a 10-week summer research program for undergraduates at the University of Minnesota Duluth, founded in 1977 by Joseph Gallian, and has sinc grown to have an outsized impact on the American combinatorics and number th community. Its long list of distinguished alumnus includes Fields medalist Manj Bhargava himself. Mehtaab Sawhney and Ashwin Sah were among the participar the Duluth REU during the summers of 2017 and 2018, respectively. In both thes years, Levent Alpöge, himself an alumni of Duluth REU from 2013, participated a research advisor.

> "Time is a river made up of events which happen, and a violent stream; for as as a thing has been seen, it is carried away, and another comes in its place, an will be carried away too."
> — Marcus Aurelius

Sah and Sawhney were both rising stars in combinatorics as undergraduates at M where they met in late 2017. Their initial acquaintance would kickstart a long an productive collaboration. During 2017 to 2024, they had produced roughly five do papers across vast areas of combinatorics such as extremal graph theory, probabi methods, and Ramsey theory. They have collaborated on at least a dozen articles were accepted at the most prestigious journals of mathematics: Annals, Inventio Duke, GAFA, JEMS. They both received the Morgan Prize in 2021 for outstandin research in mathematics as undergraduates, and finished their PhD at MIT in 20 Afterwards, Sawhney obtained the prestigious Clay Research Fellowship and joir Columbia University as a tenure-track assistant professor. Sah left mathematics, can only speculate as being disillusioned with academia, and joined D. E. Shaw & as a quantitative analyst.

On the Autumn of 2025, Sawhney would be given access to GPT-5 by Mark Sellk

it would slowly dawn on him that the system is getting increasingly better at solv
important problems in his areas of interest. Eventually, he would join OpenAI or
January 2026, to be a part of history: as a Biblical Magi witnessing the infant Chr

Successful mathematicians usually have humble beginnings. But, sooner or later,
recognize the dark undercurrents of the economics of academia, in which the pre
currency dictates much of one's career trajectory, hiring prospects, journal accep
article readership, invitation to seminars and conferences, and so on. Intrinsicall
prestige is an emergent behaviour in any creative medium: by and large, society v
and look up to as inspiration, individuals who are more creative and original thar
others. And, as in any discipline, some individuals hone the meditative practice t
allows one to obtain ever more creative ideas: the art of "catching the big fish" as
American filmmaker David Lynch likes to put it.

The trouble is, once the currency is in place, society tends to reconfigure around
rules of economics, and individuals often start valuing the prestige hierarchy mu
more than the process by which these alien ideas germinate in the human mind.
is only a flaw in said individuals insofar as every human is afflicted by Adam's cu
combination of early successes tend to exacerbate this psychological state, and
sometimes one falls into a kind of cyclical need for more prestige, often at the co
enjoyment of the creative process itself. This is best highlighted by Jacob Tsimer
own recounting of the time when he started working towards the Fields medal,
wherein he meticulously shifted his research project to align with the kind of wo
that is valued by the prestige hierarchy, maximizing his chances for the Fields m
Instances of such "perverse incentives" are abound in economic history.

I would like to posit the theory that many academics currently in AI research spa
are fundamentally misaligned by the prestige currency of academia, and the "AI
misalignment" they wish to solve isn't a bug, but a feature of their very own creat
mere reflection of their own selves projected back. During the time this essay is b
written, in mid-October, frontier AI laboratories are working to establish that th
latest batch of AI systems possess preliminary signs of "research taste" in
mathematics. The way they plan to prove it is thus: OpenAI is going to say that b

letting their internal model "Bel" run on a single problem in a particular mathematicians' research area, the model can autonomously find and exhaust all related problems in their research program.

The idea that one can strip-mine a collection of problems roughly aligned aroun single theme to establish "research taste" should at its face be laughable to any reasonable person. But to an AI researcher living in their own bubble of collectiv myopia, bound by the very paradigm of goal-aligned research that they infuse an infect these models with by reinforcement learning, taken aback by the paradigm effectiveness as gauged by various statistical measures of productivity, reaffirme their own flawed ideals of human creative and scientific processes— to them, thi unvarnished truth. At the end of the day, I would feel deeply sorry for them, had i been that they were working with a handful of enormously powerful corporation backed by large institutional and financial structures, which can act on its own a to establish new norms around aspects of human society with total impunity.

I like to imagine that Ashwin found a more fulfilling life away from academia.

VIII.
Look On My Works, Ye Mighty, And Despair!

By the end of August, boiling anger, concern and ultimately dismay within the mathematical community began bubbling up to the surface. A communal blog fo mathematicians, Proofs & Prompts, was launched earlier in the month and slowl stream of essays started arriving. Likewise, multiple essays on AI & math began cropping up under the math.HO tag on arXiv; notable among them was Max Weinreich's essay presenting a case for total opposition to the use of AI in mathematics, one that would eventually become a seed for the formation of Association for Human Mathematics in early September. Mathematicians from a the world, with wildly varying worldviews and opinions (even regarding the use o and the future of math itself) started pouring in on X, united only insofar as they a breath of fresh air compared to the opinions of the so called "techno-optimists" self-styled fashionable collective of hollow men, eternally proclaiming the same monotone plastic incantation of "more progress" in unison.

In late August, Hugo Duminil-Copin would write a Proofs & Prompts essay titlec "Care for a little more AI?", where he makes the prediction that one of the milest results in percolation theory is likely to be solved by AI soon, and how this may r the cause for celebration an average person hopes it is. He wrote (verbatim): "In r opinion, the problem lies elsewhere: the current use of AI does not empower us, petrifies us. These artificial discoveries risk decapitating entire fields before they had time to develop to their full potential. Even worse, they nuke the mathematic landscape, making it increasingly difficult to inhabit after each blast." A few days a GitHub repository managed by Anthropic and committed by Anthropic researc Justin Leder, would mysteriously appear out of nowhere in the AI terminals of ar who had asked for it, with an AI generated proof of exactly the result Duminil-C had mentioned.

In early September, six Caltech undergraduates, four of whom had previously int in tech venture capital startups like the YCombinator ecosystem, Sequoia and Neuralink, would go on to announce an AIxMath competition event "Mathathon the promotional slogan "40 hours, $2M+ AI credits, solve an open problem." The program was sponsored by OpenAI, Anthropic and several other corporations tie AI research. Soon, a coalition of current and former Caltech mathematicians pre an open letter vehemently opposing the event on September 10. Upon going live, gathered thousands of signatories within hours.

The top was starting to blow off.

Somewhere amidst the ensuing chaos, Sébastien Bubeck reached out to me on A 26 through a mutual on X, asking me to follow him so he could DM me. And so, I He messaged me in a few hours quoting a tweet I had posted. The tweet said "[O and Anthropic] are knowingly inflicting a ton of damage to the discipline of mathematics because it keeps the investor hype going", which garnered around 2 views. At the time, I had a less than 1K followers on my account. Bubeck wrote (verbatim) "If you're open to it I would like to understand your perspective better because it is definitely not my/our intention to inflict damage to math, not in the slightest, and I'm very open to update and change course. let me know if you hav

for a chat".

I admit I felt rather apprehensive after receiving this message, and couldn't at fir really quite believe a representative from OpenAI, a corporation that has practic endless political and financial power, co-founded and managed in the capacity of CEO by a weapons-grade psychopath, is personally reaching out to a random twi account with <1K followers for policy suggestions in good faith. Especially becau prior to this, I had been writing faux-antagonistic posts tagging him: for instance during the nonsofic groups fiasco, I publicly blamed him, being the math team h for playing corporate product-shipment strategy games with mathematical proof while writing the actual mathematics badly, though now I know Sawhney was m responsible for oversight on the mathematical end of things. Part of the reason fo antagonism is that it is the key to get people's attention on X, but I digress. Eithe I felt I should be very careful in choosing my words.

My initial trepidations were largely unfounded, and it turned out Sébastien indee reach out to me seeking my opinions in good faith. Nevertheless, I am glad I took time and responded professionally several hours later after careful thought. I pas here verbatim my letter to him:

> Dear Sebastien,
>
> Thank you for reaching out.
>
> I think the project your company has embarked on is fundamentally antithetic mathematics as a discipline. From private communications with a handful of l profile mathematicians, I have heard that they were extremely angry at the wa 10 problems were marketed (as a public example, see the footnotes in Fournie Facio's writeup, his interview, and try to read between the lines), and the fact t there were discrepancies between the Lean formalization and prose proofs an the prose proofs were extremely hard to read. I do not expect these to slow you down because I understand this is part of the marketing gimmick of OAI.
>
> However, if you want some suggestions:

> - On the positive end I would recommend presenting these "proofs" really as "findings", as an astronomer does by looking at the Hubble telescope, without confident claim to correctness and "settling an old problem". Most important can emphasize that maybe there are ways to generalize these findings to disco not isolated examples but conceptual phenomena, which is the ultimate goal c mathematics in my view. This way you can take the mathematical community as you speed headfirst into the nebulous "progress" into human science that y company envisions, instead of leaving behind humans and gunning for a futur where none of these theorems have any value because no one is looking at the might help to read, and re-read, Thurston's "On proof and progress in mathematics" over and over before playing the slot machine with infinite com on these new technologies of the future.
>
> - However as a cynical alternative you can just keep dumping your findings at even faster rate and larger volume without any heed to what I expect is 50% of mathematical community that are opposing this new culture, without looking the detractor's approval, completely crush them all at once, dissolve and decla academic mathematics as an exercise in antiquity so that we (the 50%) can get faster from this furnace and just become a religion; because no matter what, e superintellience arrives and has the capacity to solve all of math, some of us w math "the old way". We just want to do mathematics instead of being endlessl trapped in this artificial discourse.
>
> Anyway these are all my opinion and it's a common sentiment among people I to but I am under no pretense of being a representative of some broader math research community. I hope these are of some use. I'm a little baffled by havin be put in the position to suggest policy for OAI given my obvious and open di for the corporation, and its influence in my field, and more importantly how I quite literally a nobody. But I have nevertheless tried to collect some thoughts present them to you as such.

Sébastien responded to me saying that the reason they did a few Erdos problems earlier in the year, then unit distance problem and then the 10 Astra problems is

to update the mathematical community that these capabilities are real, and notif community ahead of schedule so they can reconfigure institutional structures. H reassured me (verbatim) "We will NOT continue to release proofs. That would be distasteful and against the principles that we want these tools to empower existi communities. we might do one here and there, if it's like interesting in some way you can rest reassured that we will NEVER do a play on "volume" and try to floo zone with a ton of proofs that nobody cares about. That would kill the field."

A few hours later, lead reasoning researcher at OpenAI, Noam Brown, reaffirme in a public tweet. He said (verbatim), "At OpenAI, we aim to devote time to findi and announcing math results from internal models only when they would meaningfully change people's understanding of the pace of AI progress. Our mai focus is shipping great models so everyone can use them to make discoveries of t own." In the replies, someone asked if they'll take a stab at a Millennium Prize problem. He responded "We already did, it didn't solve any of them. But we have thrown a ton of inference at them.", foreshadowing events that would transpire a week later.

On August 30, Sébastien would privately message me by quoting one out of a ser tweets I had made where I revealed (an information I had obtained from two sepa reliable sources) that the OpenAI math team was internally torn about what to d the vast collection of results their models had proven during benchmarking sessi had made the point that Ronen Eldan, James Lee and Sébastien Bubeck all had differing opinions, but how it does not really matter, because the corporation the employed at structurally only optimizes for profit, even if professional mathemat stands to be collateral. Sébastien told me (verbatim), "Just want you to know, the no money to be made in math, that's really not the goal. also we are giving away seats for free! (Btw if you want one I can give you a pro subscription)". Of course, declined his generous offer. A short conversation ensued where I underlined my that it doesn't truly matter what the goal is, for the final outcome to be devastati also expressed some limited skepticism about the idea that "there is no money to made in math". The conversation ended on an amicable note.

At the time, I wondered if Sébastien's offer to give me a free pro subscription wa soft bribe of sorts because my posts were getting too much attention among the commentariat. Now I see that was just paranoia on my part: the offer was well-meaning, but it comes from the same sociopathic place as OpenAI's general inte to do "good for mathematics" or more generally "good for humanity" while pract everything they touch turn to ash. Yet, somehow, their employees continue to bu this altruistic idea not realizing that it's not their intentions but the core values c corporation they work for, which is aligned towards a project that values only Ca that dictate the outcome of their actions. An outcome that, at the end of the day, tantamount to widespread disenfranchisement and alienation of individuals.

On October 12, after hanging their dirty laundry in public by turning the solutio the Navier-Stokes problem into a public spectacle of corporate competition, afte releasing a batch of poorly-vetted, not even all formalized, AI-generated solution 400 open problems in mathematics that annihilated several research programs, a paralyzing countless young mathematicians in name of "progress", after all of thi OpenAI researcher Noam Brown had found in himself the gall to respond, pushi back against a tweet by David Bessis (correctly, in my opinion) claiming that Ope does not care about the future of mathematics. His response was (verbatim): "... M of the OpenAI researchers behind these results are math PhDs or former math professors. If anything, they care too much about the future of mathematics. AI i disruptive but we also think that it will be an invaluable tool for the future of mathematics."

> ADRIAN VEIDT: The Comedian was right. Humanity's savage nature will inevitably lead to global annihilation. So in order to save this planet, I have to it... with the greatest practical joke in human history.
>
> DAN DREIBERG: Killing millions?
>
> ADRIAN VEIDT: To save billions. A necessary crime.

IX.
Do Androids Dream Of The Clay Millennium Prize?

In early September, various unsubstantiated rumors began circulating on X that Anthropic had solved "at least two" Millennium Prize problems.

On September 08, mathematician Tristan Buckmaster posted a tweet on Mastodo containing four documents: three of them contained finite-time blowup solution three separate PDE's related to fluid dynamics. The fourth document was a state released on behalf of himself and Levent Alpöge. In the statement, Buckmaster discloses that they were working on these problems along with finite-time blowu the Navier-Stokes equation, for the past year. Along the way, they had made use o several LLMs. The statement then goes on to apologize for what it's about to say and how the headline really should have been a call for the mathematical academ wake up to the capabilities of LLMs, how their findings are indicative of a "Deep Kasparov" moment in mathematics, and laments for having external reasons forc hands into changing that intended message.

In the rest of the statement, Buckmaster tells an astounding story. He recalls reac out to an *unnamed prominent mathematician* at OpenAI to clarify the provenance o works with Alpöge, which he claims were in personal capacity without any institutional involvements, and to defuse the corporate tension originating from Millennium Problem rumors, which had spilled quite a bit far from social media into the real world. On September 06, Buckmaster met with this unnamed mathematician and Sébastien Bubeck on a call, in which he was informed that in OpenAI model had produced a 100+ page proof of finite time blowup for the forc Navier-Stokes equations, settling statements C and D of the famed Clay Prize problem, with "very little human input".

Buckmaster had, then and consistently since, expressed skepticism regarding the little human input" part. According to him further chats with Sébastien revealed there was a team of humans working on this problem and an enormous amount o compute had been used (compare Noam Brown's statement a week ago "we haver thrown a ton of inference at it"). Buckmaster would go on to make insinuations t OpenAI's breakthrough was possible only after them hearing that Buckmaster w involved in the project with Alpöge. From his recent Numberphile interview

(verbatim): “Eventually I said… I sort of did a loaded question to Sébastien, I said you agree that you only started working on it after the rumors?” and he said yes.” would also go on to say that OpenAI’s unforced Euler equation paper had all the mechanisms and architectural ingredients identical to a different version of the I equation paper he and Alpöge produced but did not release yet.

In my opinion, Buckmaster’s skepticism is well-placed: I have heard from separa reliable sources that there was a fairly large team, consisting of NDA’d individua we may never hear of, working tirelessly on this, often working overnight at the OpenAI offices. Buckmaster goes on to say (verbatim):

> I asked whether the model had been trained on, or had access to, our sessions Codex, into which we had been putting all our drafts for the whole of this pro was told the model did not look up user data. I asked again, about training, an did not get an answer.
>
> Two proposals were offered to me. The first was that we post our Euler result, that OpenAI post its Navier-Stokes result the next day. The second was that, a posting Euler, I alone write a paper presenting the Navier-Stokes result, acknowledging that an internal OpenAI model had resolved it. Sebastien twic asserted that he wanted Levent removed from authorship, and said it would al simple if only it were not the case that, and it was so annoying that, Levent wo Anthropic. It was also said that if OpenAI posted after us, they would say that deserved the Clay Prize, and that we were the “closest humans to the problem declined both offers.
>
> I said that if OpenAI released its result in the way proposed I would go public what happened. The reply was, “Why would you ruin your career?” I replied th am an academic, and asked why he thought going public would ruin my caree reply was, “If you don’t want me to be nice, then I don’t have to be nice”.

I must admit that, while reading Buckmaster’s account of this extraordinary exch I was not only left flabbergasted but mildly impressed. I suppose this reaffirmed faith in OpenAI’s messianic sociopathy that was earlier defused by Sébastien’s ra

polite messages to me.

Around 12 hours after the release of Buckmaster's statement, OpenAI announced they have solved the Navier-Stokes problem. Sam Altman personally quote-tweet OpenAI's announcement tweet and said (verbatim) "one of the most amazing mo for me in OpenAI history was watching this happen over the past week". OpenA announcement said "the agents sent 2.7 million messages and used approximatel billion output tokens" for Navier-Stokes, and the successful group involved "on t order of 10000 concurrent agents", and that the result arrived about 88 hours afte first agents were launched. Several individuals estimated the API-equivalent cos roughly 20-30M USD. In an X post, Noam Brown himself confirmed (verbatim) "' this result cost millions of dollars."

Meanwhile, Sébastien Bubeck wrote a lengthy tweet pushing back against some accusations by Tristan Buckmaster. In particular, he said (verbatim):

> I was confused why one would turn an incredible source of celebration (of the achievements!) into such bickering, which is when I said that I did not unders why one would risk their career [over unfounded accusations]. Genuinely, at th moment, I was trying to care for him and do a one last ditch attempt to get a c to give them all the credit they deserve. I deeply apologize for this extremely p choice of words, it is the opposite of what I was trying to convey. (I should say retracted them on the spot by the way.)

In a response on Mastodon, Buckmaster writes (verbatim):

> Sébastien's statement below is wildly false. Not only did he never apologize, h doubled down on his threat 'if you don't want me to be nice, then i don't have nice'.
>
> Sam [Altman]'s tweet that Levent refused to communicate is also false and contradicted by Sebastien himself.
>
> Sam [Altman]'s tweet that they had a different approach is also false. It is prob best the OpenAI due their due diligence before making further claims.

Several other mathematicians rallied in the general direction of Buckmaster's co towards IP theft from OpenAI. In a Mastodon post on September 10, Andreas Th recalls an exchange he had with OpenAI after the non-sofic group announcemen which I paste here verbatim:

> I wrote in an email to Mark Sellke and Sebastien Bubeck shortly after the surp finding of a non-sofic group using the methods of Kun and myself:
>
> "Another point is that I and a colleague in Dresden were discussing the expan matching problem and various extensions of the work with Gabor Kun activel the last months with ChatGPT, so that we are of course curious if that was pa the training data or accessible to the reasoning process. There is a certain (fra unacceptable) lack of transparency here; and I fear it will damage the commur process of math more than the new AI-generated results will benefit the subje
>
> Mark Sellke's complete answer was: "Regarding your conversations with Chat that did not happen."
>
> I had explicitly asked about two different things: (1) whether our conversation entered training data, and (2) whether they were accessible to the solving proc The categorical answer now looks as though it addressed only direct access ur (2). No such qualification, explanation, or evidence was given. I take this as dishonesty to say the least.

Similar concerns were raised about Anthropic, in relation to the Jacobian Conjec In a pre-print titled "On endomorphisms of affine spaces and the Jacobian probl by Alexander Borisov, Ofer Gabber and Adrian Vasiu, where the third author ma the following statement

> Very unfortunately and without me being aware, the in-progress version of thi work dated January 14, 2025 was shared publicly at the link [...] on January 16, My understanding is that after 18 months it became a feed of the AI models.

All the disparate circumstantial evidence surround these events lead me to concl that we are probably witnessing the biggest IP theft project in the human history

perhaps unintentionally and as a consequence of alignment issues innate to the products shipped out by AI labs, and that the AI labs are desperately trying their to dismiss these concerns as either bad-faith accusations or unfounded paranoia the users, compounded by techno-optimist pundits on social media trying to undermine these issues by attacking the very concept of IP itself. The latter argu would go along the lines of the following caricature: "so what if AI has stolen you academic work without attribution, at least it'll find a cure for cancer!".

My plea to applied science industries (biotechnology and pharmaceuticals, mater science, defence contractors) and the financial sector (for instance, trading firms) are considering integrating top of the line AI models into their operations is to t the mathematicians' concerns about intellectual property theft seriously. Do you want your proprietary research, trade secrets and commercially sensitive data fre roaming around the servers of AI corporations, with little material assurance as t how that information might be stored, accessed, or used?

> "I would add a Tolkien analogy and say that Math is fighting the Battle of Hel Deep for the human role in the future of STEM right now"
> — Francesco Maggi

X.
Encounters At The End Of The World

In the midst of the Navier-Stokes chaos, past acquaintances and interns of Sébas Bubeck came out openly on X with more accusations of unscrupulous behaviour. Sébastien reached out to me again on September 10 with a casual joke. I told him speak frankly. I paste here some of the discussion verbatim:

> BUBECK: 100% honest answer: personally I would like to spend my time on th other areas I care about, which are probably not your cup of tea, but like, healthcare. if we have a good machine, accelerating medicine seems like a goc idea.
>
> NIHIL: And your corporation would botch it up exactly the way it's botched u math academia. We know this.

> BUBECK: :-(
>
> NIHIL: You have left a trail of destruction at your wake. I mean I don't blame
> But that's what it is.

At this point, Sébastien asked me for (verbatim) "One Name, One Person" that I
trust within the mathematical community to make the right decisions, whom the
mathematical end of OpenAI's explorations could be handed over so that it coul
decided from within. I thought for a long time and came up with names of four
mathematicians who could be put in a committee to decide the best practices for
mathematicians moving forward. But I struck a bargain:

> Since you solicit my suggestion, here is my concrete offer. Here is a list of
> mathematicians, including some extremely serious people, who are opposed t
> Mathathon [list of 1K+ signatories in Open Letter about the Mathathon]. In th
> you can see, for instance, that there is a person who has a 100% chance of win
> the Fields medal in 2030, if it still exists.
>
> My offer is this: withdraw funding from Mathathon as an act of goodwill. This
> show not just to me but to a large section of the community that OAI is seriou
> about preserving mathematics instead of destroying it, and that you, Sebastien
> Bubeck, are acting in good faith.
>
> If this is acceptable, I can tell you who the 4 people you should consult are. Tw
> them are American mathematicians, and will very likely be willing to talk to y

The next day, OpenAI scientist Dan Roberts announced that they are withdrawin
funding from Mathathon. I kept my end of the bargain by giving Sébastien the fo
names I had thought of. Eventually, a nine member committee called Advisory G
on Mathematics and Artificial Intelligence (AGMAI) would be formed, supporte
the Institute for Advanced Study. None of the four names I had suggested would
there, but perhaps that was expected for various reasons.

On September 29, AGMAI would announce guidelines for responsible release of
results generated by AI labs. On the top of their article, they would state clearly

(verbatim):

> At present, some frontier AI labs are testing advanced mathematical problems proprietary models that remain inaccessible to the broader scientific commun Our recommendations are formulated with this practical context in mind. Hov ideally, they would not do so. We want to state clearly from the start: we do no endorse this practice, and we ask them to stop testing advanced mathematical problems on proprietary models.

On October 06, OpenAI would release a GitHub repository containing around 4(mathematical results, some of which settled (or killed, depending on how you loc it) entire research programs that had been guiding certain fields for a decade or r OpenAI would go on to declare that they have been collaborating with AGMAI, a use that a soft-endorsement for their release.

The magnitude of impact on mathematics that this release had is hard to summa a few lines. I refer the readers to this Proofs & Prompts article to get an idea of th reactions from the broader mathematical community. Suffice to say, working mathematicians were not particularly thrilled by this development. Most of the p in the repository are unreadable to a human mathematician without repeatedly interfacing with the very AI models that created them. The task of understandin these proofs is closer to that of a cataloguer than a creative artist.

A fair share of the proofs are not formalized, which comes into question the inte AI lab epistemology regarding benchmarking. If, as some claim, these problems v indeed benchmarking problems, how is correctness of the output evaluated? Presumably, one offloads the proof into some collection of agents, with the parer system given the task to convince these agents that the proofs are correct. Upon completion of the process, it is decided that with some sufficiently high probabil the proof is correct, and then the session is shelved. This is all speculation on my gathered from private correspondence with various people; the truth is we do no know how these labs operate.

On October 07, OpenAI would retract three articles due to a critical sign error th

kills the proof strategy in each of the results. Human mathematicians pointed ou the sign error is actually rather incidental: the proof strategy itself is unsound. I like you to imagine a scenario where reward-hacked AI models would make subt errors in drug design or critical infrastructure or algorithms that, if went undete would put consumers at risk.

In such a scenario, which seems increasingly plausible in the near future, would attributed to people who underplayed "AI risk", or to the negligence of those at t helms of these powerful models? I think it would be prudent for investors in thes large corporations to exercise some caution.

OpenAI and Anthropic recently announced a contribution of $5M each to a Simo Foundation initiative to, in OpenAI's words, "help mathematicians examine, understand, and build on results produced by AI". These corporations further pr access to their models and compute-resources to researchers supported by the Si Foundation.

While limited and meagre contribution to a small section of the community in compensation for demolishing the wheels and pinions of academic mathematics, annihilating research programs that distinguished mathematicians and their stu were working on, and sowing the seed of volatility and chaos within the mathem community— could seem like a gesture of goodwill, I worry that this takes us on closer to the great institutional capture by private capital that these corporations been actively undertaking, a project that vastly overshadows that of Elsevier, and that will spill over into reshaping, and ultimately controlling, other aspects of the human society soon enough.

XI.
Epilogue

The story I have told here is a memoir: an account of events that transpired in th space between mathematics and AI, as witnessed from the limited vantage of sor that got a chance glimpse into the eye of the storm. Some of the details are so exceptionally surreal that I felt compelled to preserve them in writing, as faithful

can, in roughly the order of occurrence.

But more importantly, I wanted to illustrate that every step of the way it was the deliberate decisions from employees of these giant corporations that eventually l the rapid rupture in the values and principles that kept the mathematical commu held together. When speaking about AI, I often see people criticizing the technol AI could never do this, or that. After living through AI corporations' obsession w mathematics in the span of the last six months, I have come to realize that it's no tool that is dangerous, but the *people* who are at the helms of the tool.

It would be far too simple if the individuals involved were plain evil. It is more th that their very involvement with these large corporations — which are guided by forces of Capital beyond individual reckoning, which by means of a great churn t turns human conception of value and economic productivity upside down, effect aim to establish a new world order with these corporations themselves at the cen it all — when suddenly faced with the great responsibilities and their misplaced altruism, end up exacerbating their innate sociopathic tendencies while making decisions for certain large sections of society. Therefore, despite the claims they in their individual capacities that they merely want to empower humanity by rap shipment of these powerful new tools, in reality they are very much like the Watc of Alan Moore.

On August 28, I had asked Sébastien Bubeck: "What do you see as the future of professional mathematics? I think you have a vision of it when you mention 'wha coming'. This is a genuine question from someone far below". I think his respons (verbatim) was enlightening:

> Call me an optimist, but I think:
>
> 1. there is a chance for much faster progress. There are so so many questions th want answers to, and now we have a chance to see them in our lifetime. It wi definitely be faster paced, and not as tranquil as it used to be.
> 2. what I was personally always drawn to (and so were the masters such as Gau the "power" that a strong theorem gives you on the actual real world. Sudden

> you can do things that you couldn't before. This was a bit lost I would say in last century, but there is a chance to reclaim an extraordinary impact of matl the real world.

I never really agreed with the point of view that we, collectively as mathematicia want "answers to questions": it had always seemed to me that the confusion *is* th point, the recognition of how much we do not know, and the process of fumbling around in the dark. Even after the AI revolution, I stand by this perspective: for e question we obtain an answer to, I think it would be prudent to ask ourselves if t questions we had asked were really the right questions we meant to ask in the fir place. To me, mathematics will always be a raid on the inarticulate: a blind and il defined search for things that may or may not exist just on the periphery of the v "with shabby equipments always deteriorating in the general mess of imprecisio feeling" (T. S. Eliot, "Four Quartets").

Sébastien's second point intrigues me the most. At the time, I had thought that perhaps he is speaking of the great decoupling of mathematics and natural scien that had happened a century ago, and maybe to an extent he did! But after so ma weeks what catches my eye now is the word "power". In the six-part documentar "Can't Get You Out Of My Head", English journalist Adam Curtis frames the dee history of artificial intelligence as follows:

> The laws of human thought that George Boole had created would become the central structure of all thinking machines, computers, because it fitted perfec with the binary switching systems inside them, either zero or one, and it was u by the machine to create endless branching pathways of binary logic, called algorithms. Out of that was going to come the dream of artificial intelligence: machines that could think independently, that could then order and manage t world as a rational system, not driven by the dangerous ideologies of the past.
>
> But back in the 1960s, as the engineers began to build the first neural network what they had forgotten was that the system of thought they were creating ins the machines did have its own history. That it had been born out of a time wh science had become deeply involved with questions of Power and Control in t

> British Empire. That what lay behind the computer logic was the aim of simpl human thought, which would finally allow you to colonize the last free outpos human mind. But unlike the old empires, where power was visible, this power would be hidden in remote places — in the servers.

Created as a result of an accident at a US government particle physics and weapc research base in New Mexico, Dr. Manhattan is a nigh-omnipotent and nigh-omniscient being in *Watchmen* who grows increasingly distant and apathetic tow humanity, primarily due to his omniscience and nonlinear perception of time. Du the course of events in the graphic novel, he is used by the US government as a weapon and eventually manipulated by Ozymandias to go along with his destruc plan.

Dr. Manhattan is portrayed as a detached and indifferent god who does not see h as slighted by the humans who weave a web around him and complies unbothere with the events that unfold, as all of it is already visible to him ahead of time, like clear stream of water:

> "Why does my perception of time distress you? Everything is preordained, eve responses. We're all puppets, Laurie. I'm just a puppet who can see the strings

The jury is still out on whether or not AI will prove to be one of the consequentia discoveries made by humanity to date. In concluding this essay, I instead urge th readers to consider the possibility that, in the hands of for-profit corporations an governments hungry for power and control, administered by flawed men that prc altruism but betray a sociopathic indifference to human consequences, it become nothing more than a silicon Leviathan: a vast and monstrous centralizing power will eat through centuries of human values, intellectual traditions and, most imp of all, the human freedom of thought.

> "The ultimate hidden truth of the world is that it is something we make, and c just as easily make differently"
> — David Graeber (1961-2020)